

DATA ANALYTICS REPORT

Employee Workforce Data Cleaning & Analysis

A Full Pipeline: From Raw, Messy Records to Actionable HR Insights

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Dataset: Messy Employee Dataset (1,020 records)
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1. Executive Summary

This report documents the complete data analytics engagement carried out on the client's employee workforce dataset. The dataset, as originally supplied, was in a deliberately messy and inconsistent state — a common real-world condition that prevents direct use in dashboards, reports, or decision-making. Our team carried out a structured, end-to-end process: importing the raw data, profiling and diagnosing every quality issue, systematically cleaning and restructuring the dataset, loading the finalized data into a relational database, and finally producing a series of in-depth visual analyses to surface workforce insights.

The objective of this engagement was twofold: first, to transform an unusable, error-ridden spreadsheet into a clean, analysis-ready dataset; and second, to extract meaningful, decision-relevant insights from that data regarding employee performance, compensation, remote work patterns, departmental and regional distribution, and hiring trends over time.

Engagement Highlights

- Processed 1,020 employee records across 12 original data fields.
- Identified and resolved missing values in the Age field (211 records) and Salary field (24 records).
- Split a combined “Department-Region” text field into two clean, independent analytical columns.
- Consolidated employee first and last names into a single, repositioned full-name field.
- Removed personally identifiable information (Email, Phone) that was not needed for workforce analytics.
- Converted categorical performance ratings into a numeric scale for statistical analysis, then restored readable labels for final reporting.
- Loaded the final, clean dataset into a PostgreSQL database table for downstream use.
- Produced 13 detailed visualizations covering performance, salary, remote work, regional distribution, and hiring trends.

The sections that follow walk through this entire process step by step, with screenshots of the actual working outputs, so that a non-technical reader can follow exactly what was done, why it was necessary, and what it revealed about the workforce. The report concludes with a consolidated set of business recommendations drawn directly from the analysis.

2. Project Scope & Methodology

The engagement followed a standard, industry-proven data analytics workflow consisting of five distinct phases:

1. **Data Acquisition** — importing the raw dataset into the analysis environment.
2. **Data Profiling** — systematically inspecting the dataset to identify quality issues (missing values, inconsistent formatting, combined fields, irrelevant columns).
3. **Data Cleaning & Transformation** — correcting, restructuring, and enriching the data so it is consistent and analysis-ready.
4. **Data Loading** — persisting the cleaned dataset into a structured database for reliable, repeatable downstream access.
5. **Exploratory Analysis & Visualization** — building targeted charts to answer specific business questions about the workforce.

The toolset used for this engagement consisted of Python with the pandas library for data manipulation, the matplotlib and seaborn libraries for visualization, and PostgreSQL as the destination database, connected via SQLAlchemy. This is a widely-used, industry-standard combination chosen for its transparency, reproducibility, and auditability — every transformation step is recorded in code and can be re-run or verified at any time.

3. Data Cleaning & Preparation

Before any meaningful analysis can take place, raw data must be inspected and cleaned. This is frequently the most time-consuming part of any analytics project, and it is also the most important: insights drawn from messy or inconsistent data cannot be trusted. This section documents every cleaning decision made, in the order it was performed, along with the reasoning behind each step.

3.1 Initial Data Inspection

The raw dataset was loaded into the analysis environment and immediately profiled to understand its size, structure, and quality. The dataset contained 1,020 employee records spread across 12 original columns: Employee ID, First Name, Last Name, Age, Department-Region (a combined field), Status, Join Date, Salary, Email, Phone, Performance Score, and Remote Work.

Check Performed	Result
Total records (rows)	1,020
Total original columns	12
Missing values — Age	211 records (20.7%)
Missing values — Salary	24 records (2.4%)
Missing values — all other fields	0 (fully populated)
Duplicate column-naming style	Mixed case, spaces (e.g. "Department_Region")

Table 1: Summary of initial data quality checks performed on the raw dataset.

This initial profiling immediately revealed the dataset's nickname was well-earned: roughly one in five employees had no recorded age, a smaller but still meaningful group had no recorded salary, and several columns combined two distinct pieces of information into one (department and region) or contained personal contact information that added no analytical value while introducing unnecessary privacy exposure.

3.2 Standardizing Column Names

All column headers were converted to lowercase and spaces were replaced with underscores (for example, "Department_Region" became "department_region"). This is a standard best practice: it prevents case-sensitivity errors during analysis and ensures the column names are compatible with database systems, which are typically case-sensitive and do not handle spaces well.

3.3 Handling Missing Age Values

211 employees (just over 20% of the dataset) had no age recorded. Simply deleting these records would have discarded a fifth of the workforce data — an unacceptable loss of information. Instead, the missing values were filled using the median age of the remaining employees.

The median was deliberately chosen over the average (mean) because the median is far less sensitive to extreme values or skewed distributions, making it a safer, more representative estimate for filling gaps. The median age across the dataset was found to be 30 years.

Age	Count Before Cleaning	Count After Cleaning
25	206	206
30	205	416
35	188	188
40	210	210
Missing (NaN)	211	0

Table 2: Age value distribution before and after median imputation. The 211 missing records were all assigned the median value of 30, which is why the count for age 30 rose from 205 to 416.

After filling the gaps, the age column — which had been stored as a decimal number (e.g. 25.0) because of the missing values — was converted to a clean whole-number (integer) format, since age is naturally a whole number.

3.4 Creating an Age Group Segmentation

To make age-based analysis easier and more business-friendly, employees were grouped into descriptive age bands — Young-Adult, Adult, and Middle-Aged — using a quartile-based

segmentation method. This converts a continuous numeric field into clear, interpretable categories that are far easier for stakeholders to reason about than raw ages.

Age	Assigned Age Group
25	Young-Adult
30	Young-Adult
35	Adult
40	Middle-Aged

Table 3: Sample mapping of age values to age group categories.

3.5 Splitting the Combined Department-Region Field

One of the more structurally significant issues in the raw data was that department and region were merged into a single text field, separated by a hyphen (for example, “DevOps-California”). This made it impossible to filter, group, or chart by department and region independently. The field was split into two separate, clean columns — department and region — and the original combined column was removed.

Original Combined Value	department	region
DevOps-California	DevOps	California
Finance-Texas	Finance	Texas
Admin-Nevada	Admin	Nevada
Cloud Tech-Florida	Cloud Tech	Florida

Table 4: Example of the department_region field being split into two independent columns.

This single transformation unlocked all of the department-level and region-level analysis presented later in this report.

3.6 Consolidating Employee Names

First Name and Last Name were stored as two separate columns. These were merged into a single, more usable employee_name field, and the original two columns were dropped. The new field was then repositioned to sit immediately after the Employee ID column, both for readability and to match standard reporting conventions where identifying information appears first.

3.7 Handling Missing Salary Values

24 employees (2.4% of the dataset) had no recorded salary. Rather than filling these gaps with a single blanket value — which would ignore real pay differences across departments and regions — each missing salary was filled using the median salary for that specific employee's department and region combination. This is a more precise, context-aware imputation method: an employee

in a high-paying department and region is filled with a value representative of their actual peer group, not a generic company-wide average.

After imputation, salary values (originally stored with decimal cents) were rounded to the nearest whole dollar and converted to a clean integer format, which is standard practice for salary reporting and avoids implying false precision.

3.8 Removing Non-Analytical, Privacy-Sensitive Fields

The Email and Phone columns were removed from the dataset entirely. These fields contained personally identifiable information (PII) that served no analytical purpose for workforce performance, compensation, or HR planning. Removing them reduces unnecessary privacy and compliance exposure while keeping the dataset focused on the information that actually drives business decisions.

3.9 Standardizing the Performance Score Field

The performance_score field originally contained four text categories: “Poor,” “Average,” “Good,” and “Excellent.” To allow statistical operations such as averaging performance across groups, these categories were temporarily mapped to a numeric scale (Poor = 1, Average = 2, Good = 3, Excellent = 4). This numeric version was used for calculations such as average performance score by age group. Once the necessary statistical aggregation was complete, the scores were mapped back to their original, readable text labels for all final reporting and visualization, ensuring charts remain intuitive for business stakeholders rather than displaying unexplained numbers.

3.10 Final Column Structure

After all cleaning steps, the dataset was reorganized into the following final, analysis-ready structure:

#	Final Column	Description
1	employee_id	Unique identifier for each employee
2	employee_name	Full name (merged from first and last name)
3	age	Employee age in whole years (missing values imputed)
4	age_group	Categorical age band: Young-Adult / Adult / Middle-Aged
5	status	Employment status: Active / Pending / Inactive
6	join_date	Date the employee joined the company
7	salary	Annual salary in whole dollars (missing values imputed)
8	performance_score	Performance rating: Poor / Average / Good / Excellent
9	remote_work	Whether the employee works remotely (True/False)
10	department	Department the employee belongs to

#	Final Column	Description
11	region	Geographic region the employee is based in

Table 5: Final, cleaned column structure used for all subsequent analysis.

Why This Matters

- Every missing value was resolved using a statistically sound, defensible method (median imputation, and group-aware median imputation) rather than simply discarding incomplete rows — preserving the full sample size of 1,020 employees.
- Splitting and merging fields (department/region, first/last name) transformed unusable combined text into clean, independently analyzable columns.
- Removing email and phone numbers reduced unnecessary privacy risk without any loss of analytical value.
- The dataset is now fully structured, fully populated, and ready for reliable, repeatable analysis and reporting.

3.11 Loading the Clean Data into a Database

Once cleaning was complete, the finalized dataset was loaded into a PostgreSQL relational database, into a table named `employee_details`. This step is important for any client wanting to use this data beyond a single one-off analysis: storing the data in a proper database (rather than leaving it in a spreadsheet or notebook) makes it queryable, shareable across teams and tools, easy to keep secure and access-controlled, and a reliable single source of truth for future reporting, dashboards, or integration with other business systems.

4. Visual Analysis & Workforce Insights

With a clean, reliable dataset in place, the analysis moved to exploring what the data actually reveals about the workforce. Thirteen visualizations were produced, each targeting a specific business question — ranging from how performance varies by department and region, to how compensation compares between remote and onsite staff, to how hiring activity has trended over recent years. Each chart below is presented as an exact screenshot of the actual analytical output, accompanied by a detailed explanation of what it shows and why it matters.

4.1 Performance Distribution by Department

This chart breaks down the number of employees in each department by their performance rating (Poor, Average, Good, Excellent), making it possible to spot departments that are performing unusually well or unusually poorly relative to others.

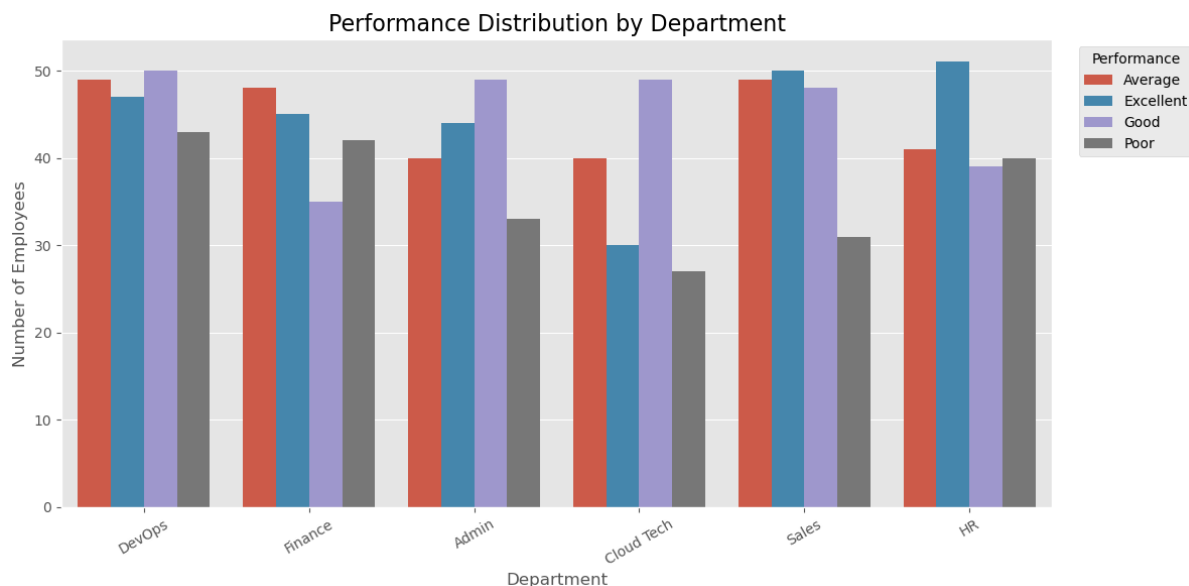


Figure 1: Performance Distribution by Department — employee counts segmented by performance rating, grouped by department.

Sales and DevOps show the strongest “Excellent” ratings among the six departments, while Cloud Tech stands out with a noticeably lower count of top performers and a comparatively higher concentration in the “Average” band. HR carries the highest raw count of “Excellent” performers of any department, but also one of the larger “Poor” counts, suggesting a wider performance spread within that team rather than a uniformly strong or weak group. Admin and Finance sit in a fairly balanced middle position across all four rating tiers.

4.2 Performance Distribution by Region

This view shifts the same type of analysis from department to geography, showing how performance ratings are distributed across the six regions in which the company operates.

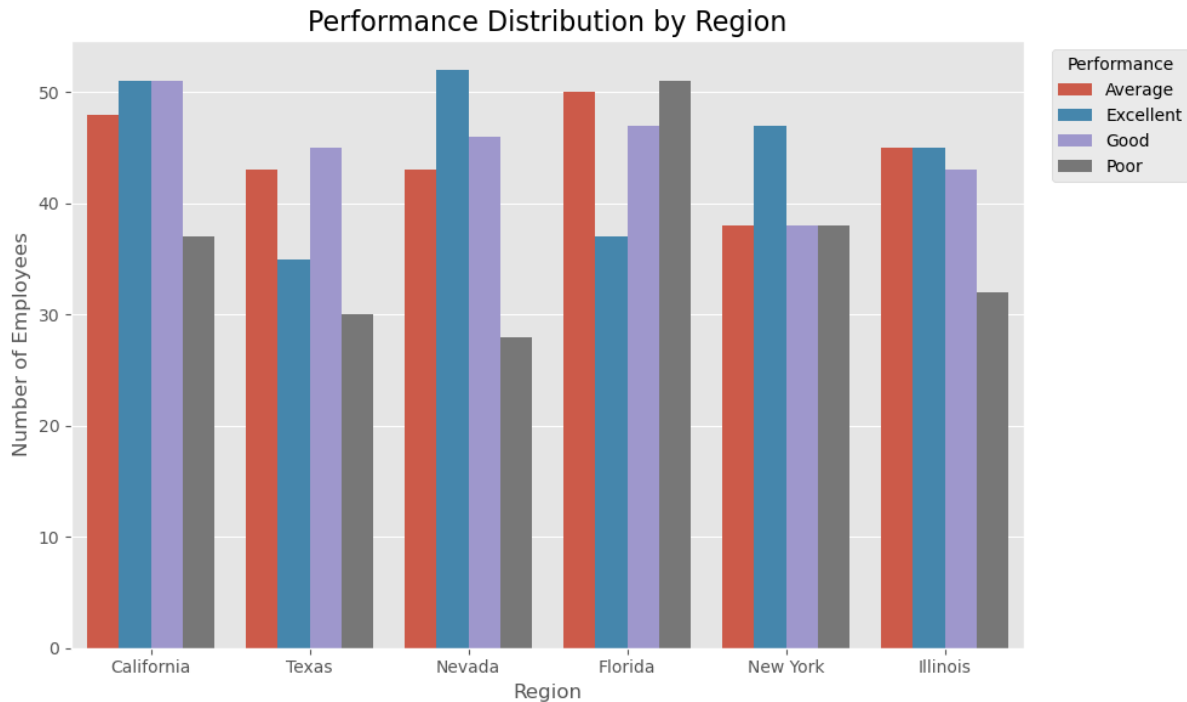


Figure 2: Performance Distribution by Region — employee counts segmented by performance rating, grouped by region.

Nevada has the highest count of “Excellent” performers of any region, while Florida shows the highest count of “Poor” ratings alongside a comparatively low “Excellent” count — making it the region most worth a closer management review. California and New York show fairly balanced, healthy distributions across all four performance tiers, while Texas has the lowest overall “Poor” count, a positive signal for that region's workforce quality.

4.3 Employee Distribution by Department and Region

This heatmap cross-tabulates department against region, with each cell showing the exact headcount for that department-region combination. Darker cells indicate a larger concentration of employees; lighter cells indicate fewer.

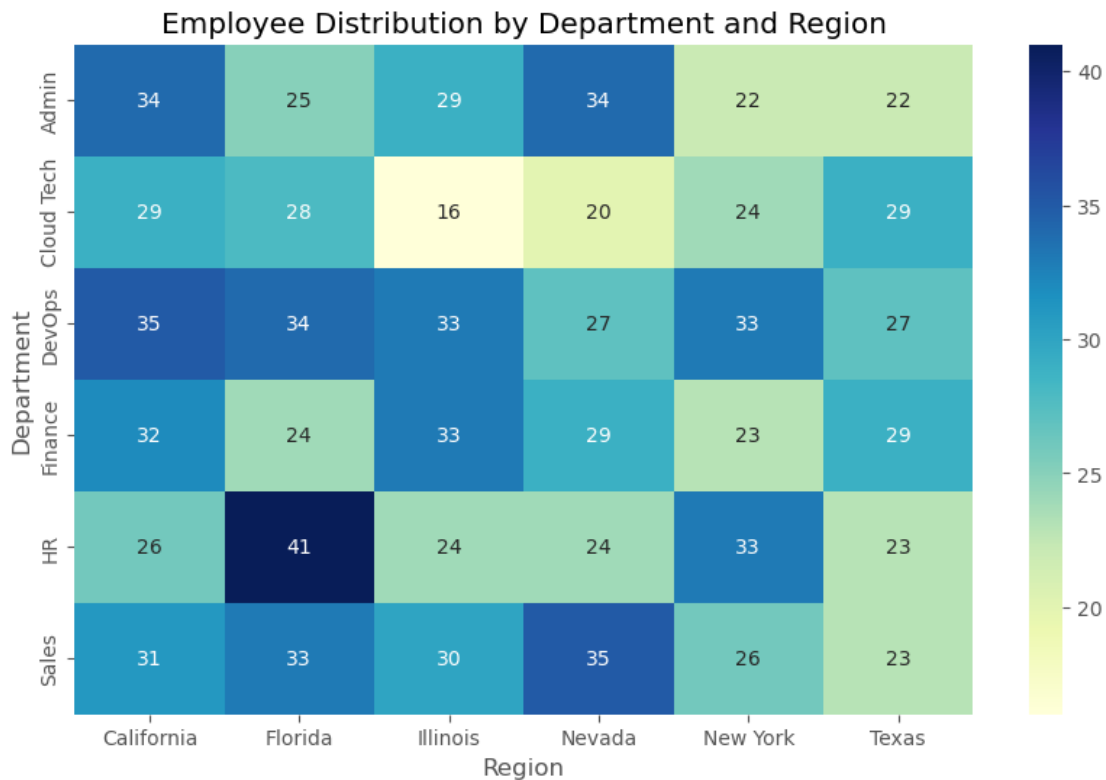


Figure 3: Employee Distribution by Department and Region — darker shading indicates higher headcount.

The single largest concentration in the entire workforce is HR staff based in Florida, with 41 employees — visibly the darkest cell on the heatmap. Several other meaningful clusters stand out: DevOps in California (35), Admin in both California and Nevada (34 each), and Sales in Nevada (35). At the other end, Cloud Tech in Illinois has the smallest headcount of any combination, at just 16 employees. This view is particularly useful for workforce planning — it quickly shows the client which office locations are heavily staffed in a given function and which combinations may be under-resourced.

4.4 Average Performance Score by Age Group

Using the numeric performance scale described earlier (Poor = 1 through Excellent = 4), this chart calculates and compares the average performance score for each age group.

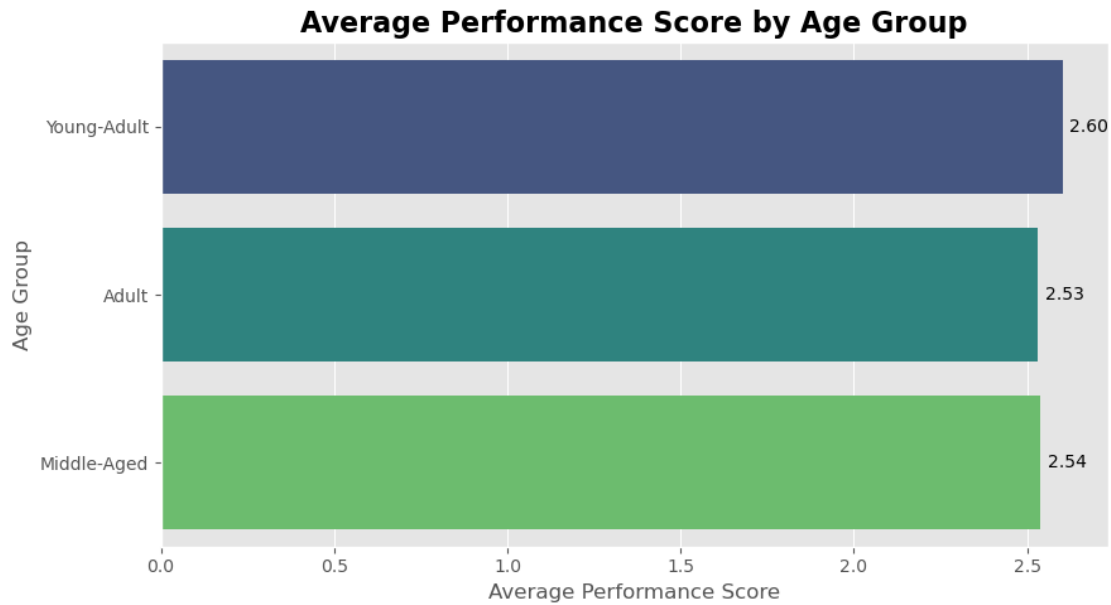


Figure 4: Average Performance Score by Age Group — scored on a 1 (Poor) to 4 (Excellent) scale.

The differences here are small but consistent: the Young-Adult group has the highest average performance score (2.60), narrowly ahead of Middle-Aged employees (2.54) and Adult employees (2.53). In practical terms, performance is fairly evenly spread across age groups in this workforce — there is no strong evidence that any one age band substantially outperforms the others. This is a reassuring finding from a fairness and workforce-planning perspective: performance appears to be driven by factors other than age.

4.5 Remote vs. Onsite Employees by Department, Across Regions

This faceted chart breaks the workforce down by region (each panel is one region), and within each panel compares the number of remote versus onsite employees for every department. This is the most granular view in the report and is designed to answer a specific operational question: where, exactly, is remote work concentrated?

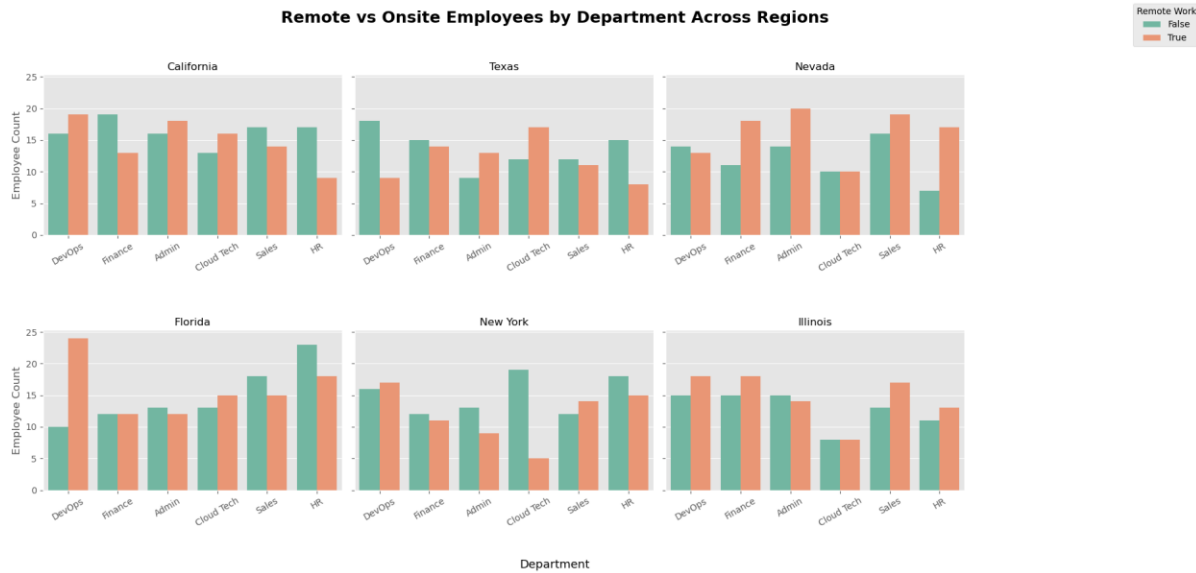


Figure 5: Remote vs. Onsite Employees by Department, broken out across all six regions.

Remote work patterns vary considerably by location. Florida's DevOps team is the most remote-skewed group in the entire company, with 24 remote employees versus only 10 onsite. New York's Cloud Tech team shows the opposite extreme — just 5 remote employees against 19 onsite, making it the most office-centric team in the dataset. Most other department-region combinations show a closer, more balanced split between remote and onsite staff. This view is valuable for facilities and real-estate planning, as well as for understanding which teams may need different management or collaboration approaches based on their work mode.

4.6 Average Salary by Department and Region

This heatmap shows the average salary for every department-region combination, allowing direct comparison of pay levels across both dimensions simultaneously. Darker cells represent higher average pay.

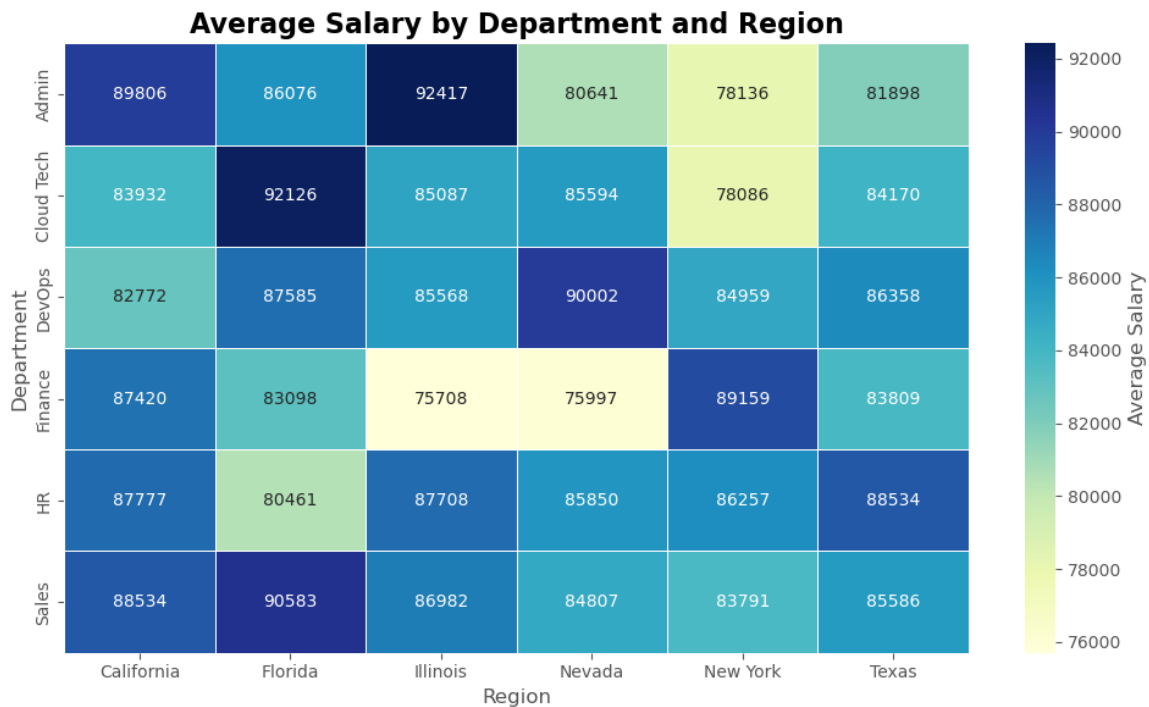


Figure 6: Average Salary by Department and Region (in US dollars).

The highest average salary in the dataset belongs to Admin staff in Illinois, at \$92,417, closely followed by Cloud Tech in Florida at \$92,126. At the lower end, Finance employees in Illinois and Nevada average roughly \$75,700–\$76,000, the lowest figures on the heatmap. Notably, Admin — often assumed to be a lower-paying function — actually commands some of the highest regional averages in this dataset, particularly in Illinois and Nevada. This kind of department-by-region pay comparison is exactly the sort of insight that supports compensation benchmarking and helps identify potential pay-equity questions worth investigating further.

4.7 Salary Distribution: Remote vs. Onsite Employees

This box plot compares the full salary distribution — not just the average, but the spread — between remote and onsite employees. The horizontal line inside each box marks the median salary; the box itself spans the middle 50% of salaries (25th to 75th percentile), and the vertical lines (whiskers) show the overall range.

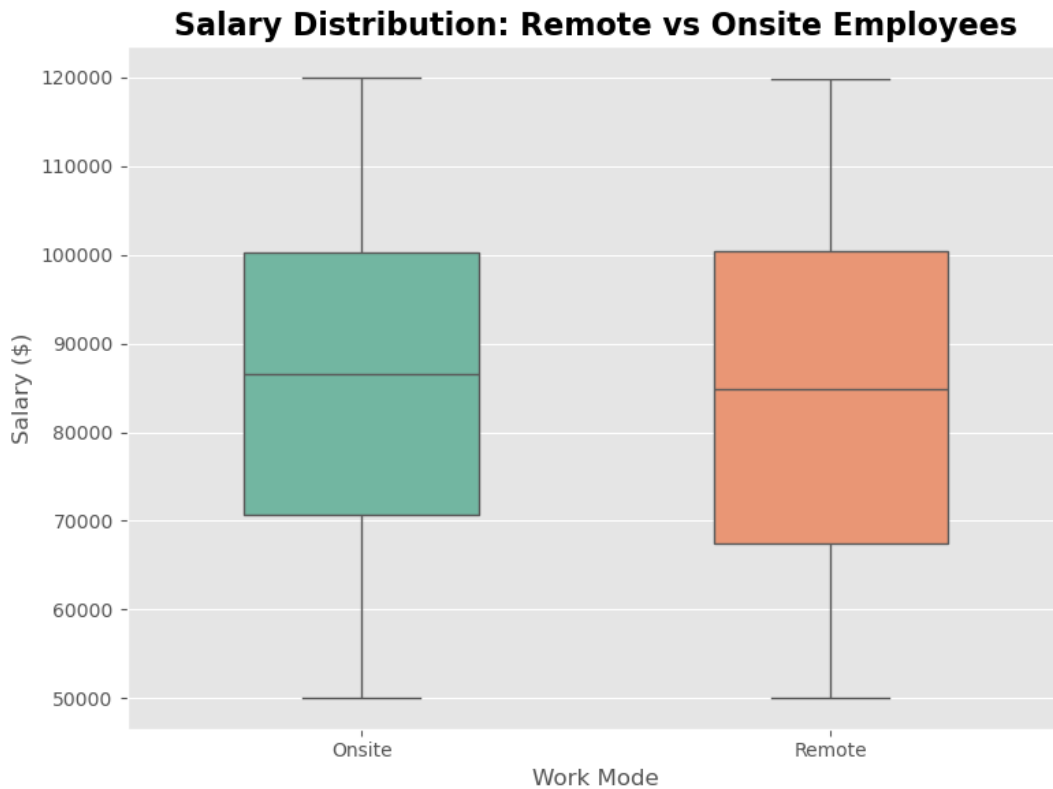


Figure 7: Salary Distribution comparing Onsite employees to Remote employees.

Onsite employees show a slightly higher median salary (approximately \$86,500) compared to remote employees (approximately \$85,000). The overall salary ranges for both groups are nearly identical — both run from about \$50,000 to \$120,000 — and the middle 50% of each group occupies a very similar band. In short, work location (remote vs. onsite) does not appear to be a major driver of pay differences in this workforce; salary is being determined by other factors such as department, region, or role rather than by work mode.

4.8 Employee Distribution by Department

This chart presents a simple but essential headcount view: how many employees work in each department company-wide, ranked from largest to smallest.

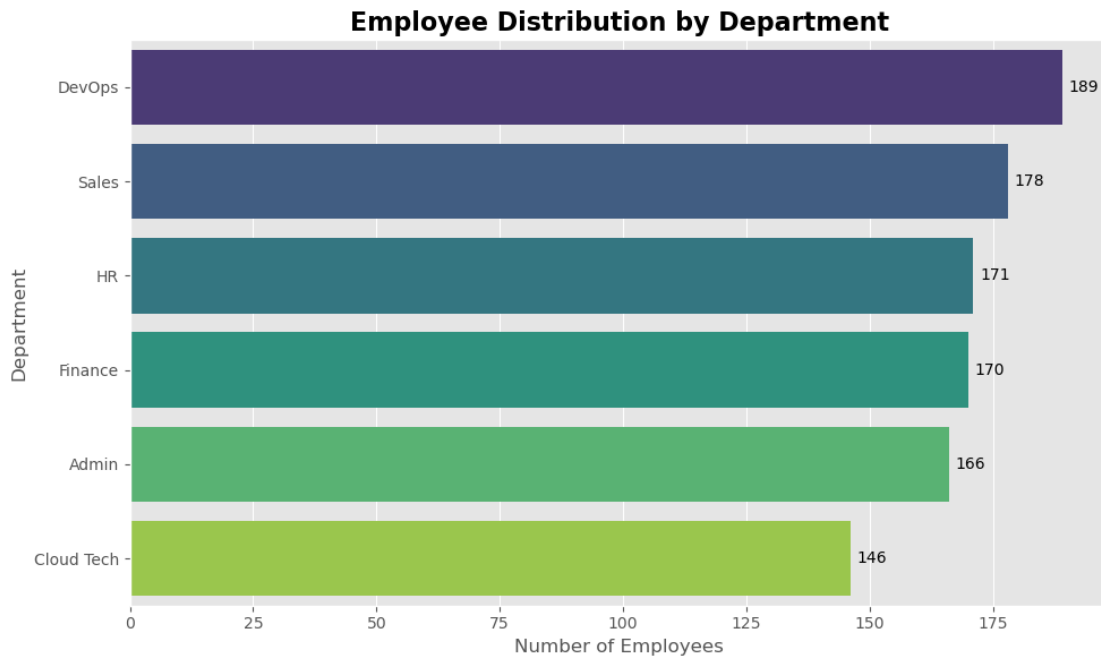


Figure 8: Employee Distribution by Department, ranked by headcount.

DevOps is the largest department in the company with 189 employees, followed by Sales (178), HR (171), Finance (170), and Admin (166). Cloud Tech is the smallest department at 146 employees — a gap of 43 employees (roughly 23%) compared to the largest department, DevOps. This baseline headcount view is foundational context for interpreting all of the other department-level charts in this report, since departments with more employees will naturally tend to show larger raw counts in other breakdowns.

4.9 Employee Distribution by Region

Mirroring the department view, this chart shows the total headcount in each of the company's six operating regions.

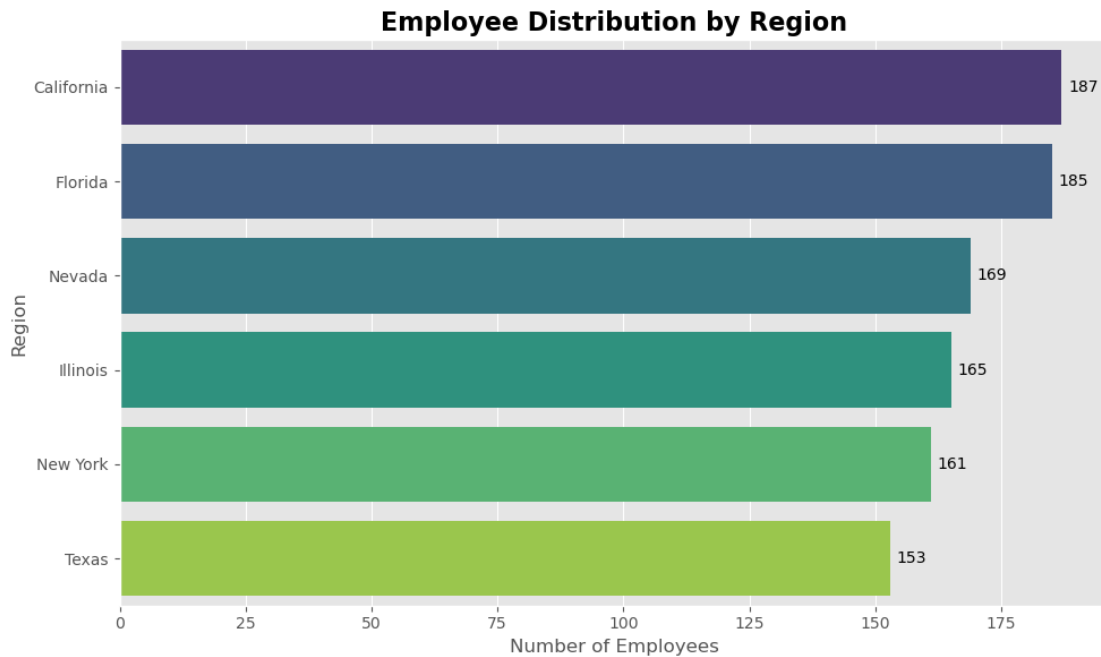


Figure 9: Employee Distribution by Region, ranked by headcount.

California has the largest regional workforce at 187 employees, with Florida close behind at 185. Texas has the smallest regional headcount at 153 employees. Overall, the distribution across regions is reasonably balanced — the gap between the largest and smallest region (34 employees, about 18%) is proportionally similar to the spread seen across departments, indicating the company has not concentrated its workforce too heavily in any single location.

4.10 Employee Status Distribution

This chart shows how the workforce breaks down by employment status: Active, Pending, and Inactive.

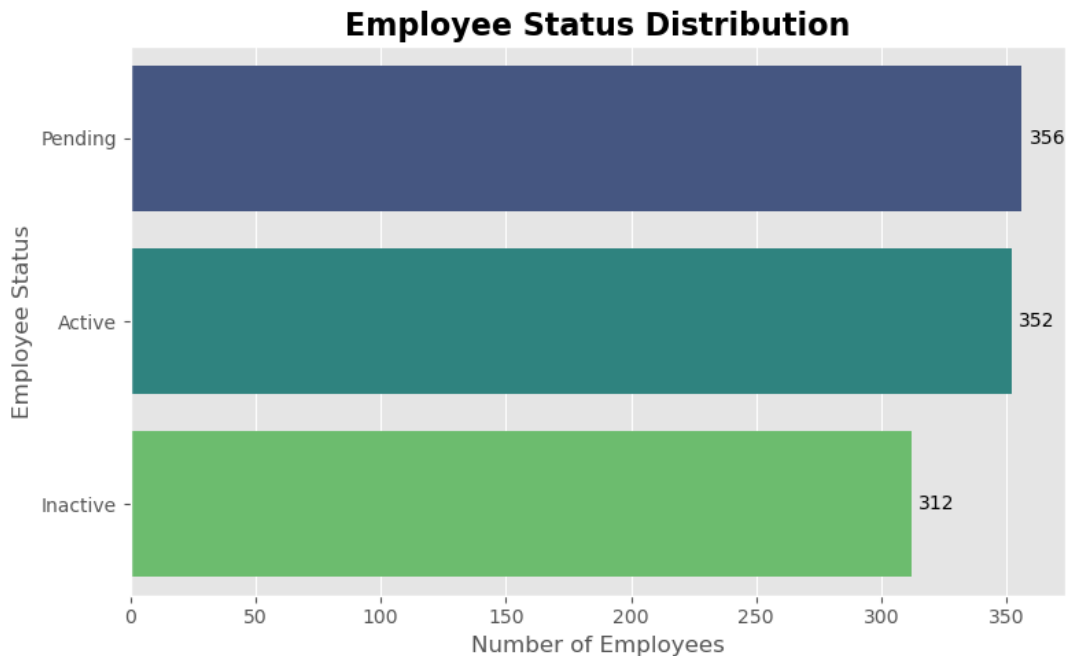


Figure 10: Employee Status Distribution across the full workforce.

Pending status is, somewhat surprisingly, the single largest group at 356 employees, just ahead of Active at 352. Inactive employees number 312. The fact that Pending and Active are nearly the same size (a difference of only 4 employees) and that Pending is technically the largest group is worth flagging for the client's HR team — a high volume of “Pending” records may indicate onboarding bottlenecks, delayed status updates, or a data-entry pattern worth investigating outside the scope of this report.

4.11 Salary Distribution by Performance Rating

This box plot examines whether higher-performing employees are compensated more than lower-performing employees, by comparing the full salary distribution across all four performance tiers.

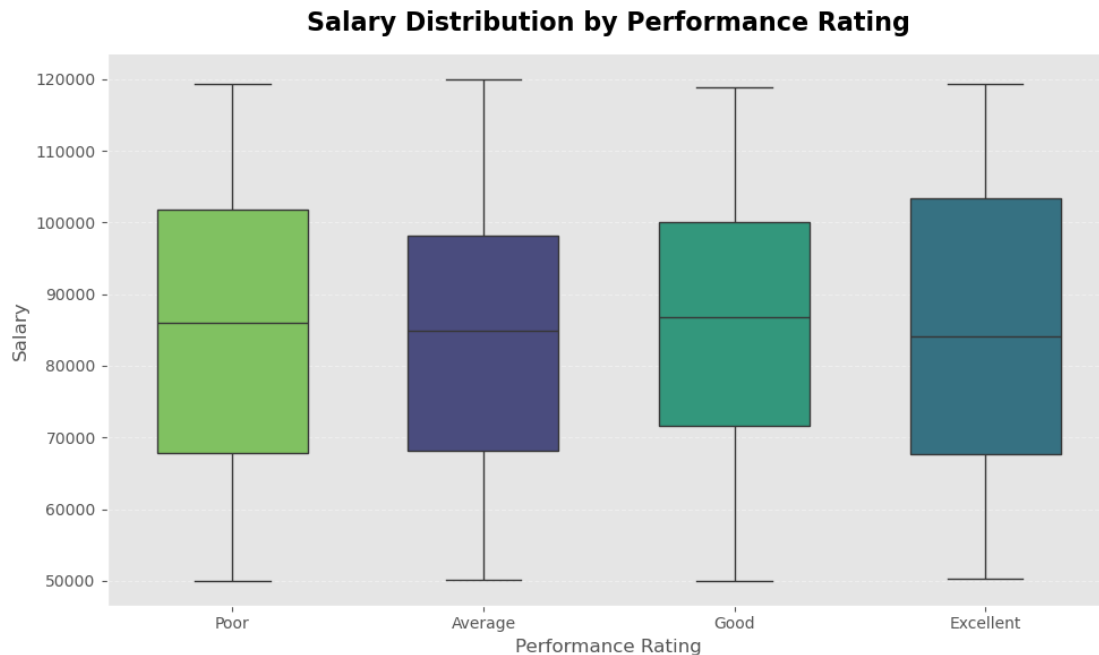


Figure 11: Salary Distribution by Performance Rating (Poor, Average, Good, Excellent).

The median salaries across all four performance tiers are remarkably close — all fall within roughly \$84,000 to \$87,000, with “Good” performers showing a very slightly higher median than “Excellent” performers. The overall salary ranges and middle-50% spreads are also very similar across tiers. This is an important and somewhat counter-intuitive finding: in this dataset, pay does not appear to scale with performance rating. Employees rated “Excellent” are not, on average, earning meaningfully more than employees rated “Poor.” This is a finding the client may want to investigate further, as it could indicate that compensation is driven primarily by role, department, or tenure rather than by individual performance outcomes.

4.12 Overall Performance Rating Distribution

This chart shows the total number of employees falling into each of the four performance categories, company-wide.

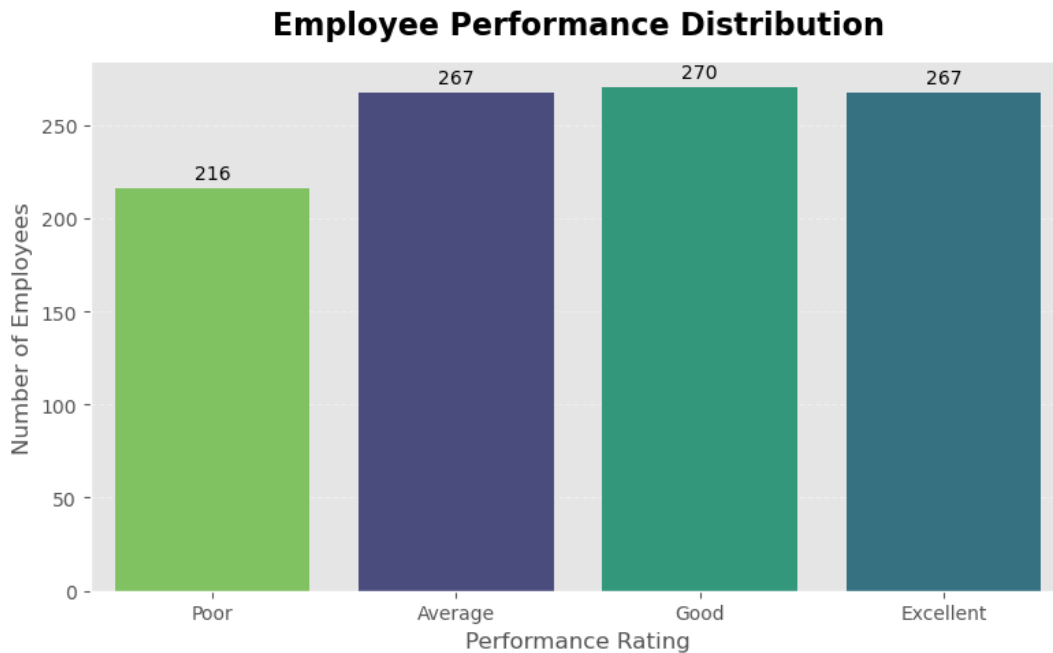


Figure 12: Employee Performance Distribution across the entire workforce.

Performance ratings are fairly evenly spread across the workforce: “Good” is the most common rating at 270 employees, followed closely by “Average” and “Excellent” (267 each), with “Poor” the smallest group at 216 employees. The relatively even spread across all four tiers — rather than a heavy skew toward any one rating — suggests the performance evaluation process is producing a reasonably balanced distribution, which is generally a healthy sign for the credibility of a performance rating system.

4.13 Performance Distribution by Work Mode (Remote vs. Onsite)

This stacked percentage chart compares the proportion of each performance rating between onsite and remote employees, making it possible to see whether work mode has any relationship with performance outcomes.

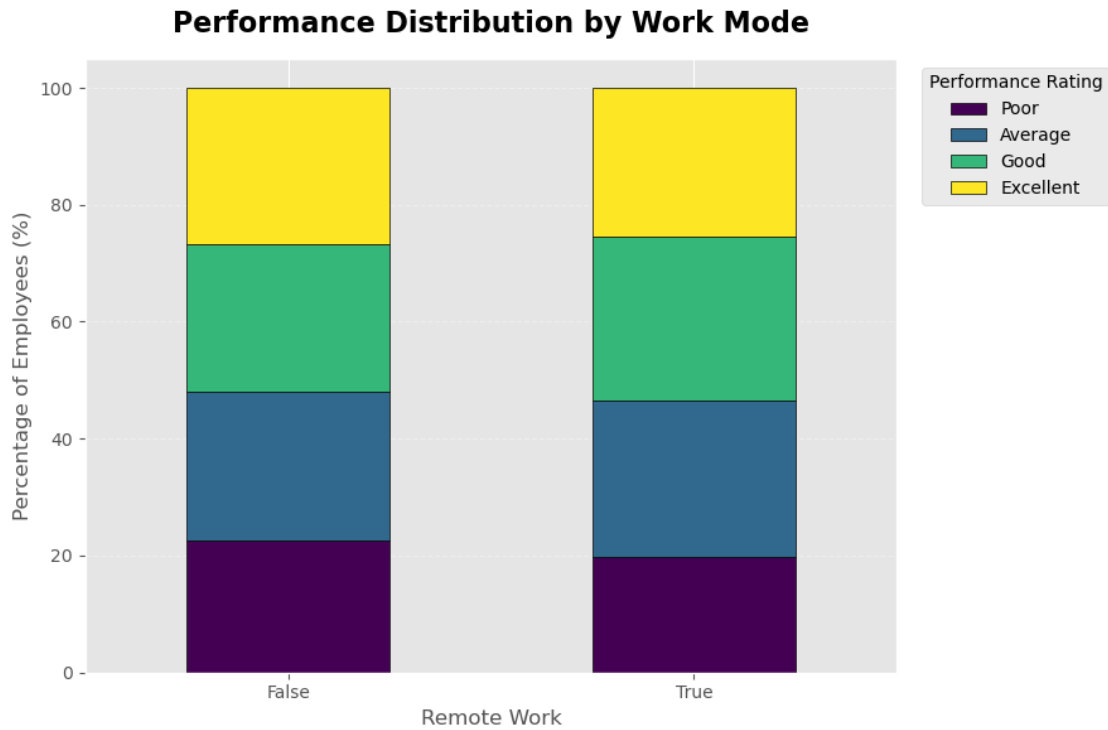


Figure 13: Performance Distribution by Work Mode, shown as a percentage of each group.

The two groups look almost identical. Onsite employees show about 23% in the “Poor” tier versus about 20% for remote employees — a small difference — while the “Average,” “Good,” and “Excellent” proportions are nearly indistinguishable between the two groups. This is a reassuring result for any organization with flexible work policies: it provides no evidence that remote work is associated with worse (or better) performance outcomes in this workforce.

4.14 Employee Hiring Trend Over Time

This line chart tracks the number of new employees hired in each calendar year, based on join date, with the peak and trough years specifically highlighted.

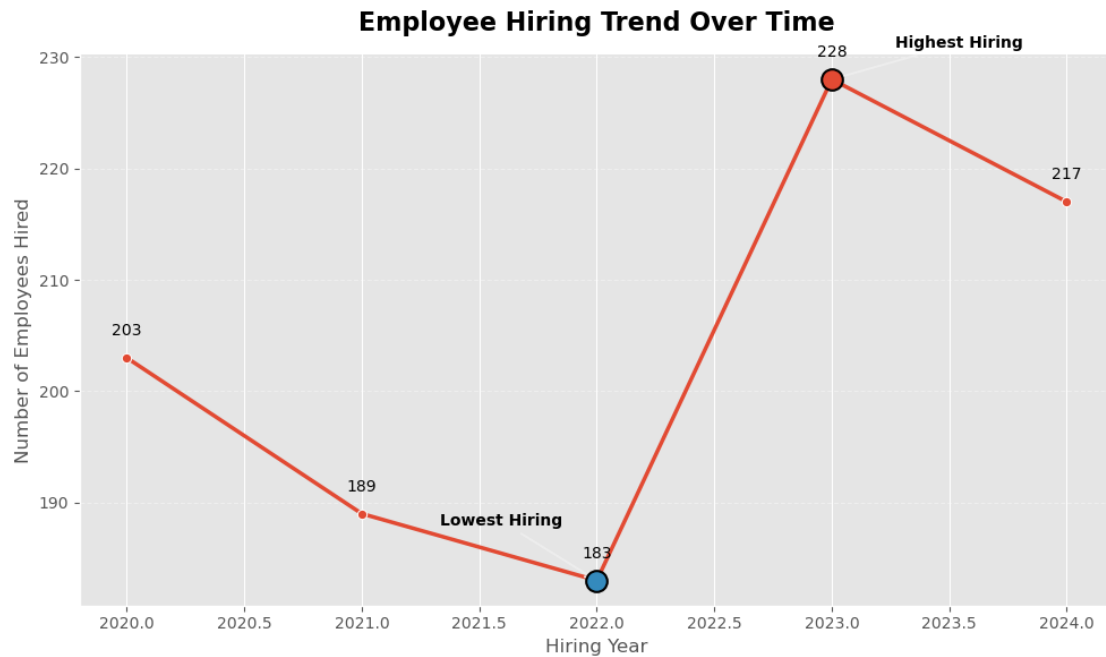


Figure 14: Employee Hiring Trend Over Time, with the highest and lowest hiring years marked.

Hiring activity was 203 employees in 2020, dipping to 189 in 2021 and reaching its lowest point of 183 in 2022. The company then experienced a sharp hiring surge in 2023, bringing in 228 new employees — the highest figure in the dataset — before easing slightly to 217 in 2024. This pattern of a multi-year slowdown followed by a sharp rebound is consistent with a company recovering from a constrained hiring period and then scaling back up. The client may wish to examine what specific business decisions or market conditions drove the 2023 hiring surge, both to understand its sustainability and to inform future workforce planning.

5. Consolidated Findings & Recommendations

Drawing together the data cleaning work and the full set of visual analyses, the following points represent the most actionable takeaways from this engagement.

5.1 Key Findings

- Data quality was significantly improved: all missing age and salary values were resolved using statistically sound, group-aware imputation methods, with zero records discarded.
- Performance ratings are broadly evenly distributed across departments, regions, age groups, and work modes — there is no strong evidence that age or remote/onsite status drives performance outcomes.
- Compensation does not appear to scale with individual performance rating; median salaries are very similar across Poor, Average, Good, and Excellent performers.
- Pay varies more by department and region than by any other factor examined — Admin staff in Illinois and Cloud Tech staff in Florida are the two highest-paid department-region combinations in the company.
- Remote work adoption varies widely by location and department, from heavily remote (Florida DevOps) to heavily onsite (New York Cloud Tech), rather than following a single company-wide pattern.
- Hiring activity dipped from 2020 to 2022 before surging in 2023, the single highest-hiring year in the dataset.
- “Pending” is the single largest employment status category, marginally ahead of “Active,” which may warrant a closer look at onboarding or status-tracking processes.

5.2 Recommendations

1. Investigate the disconnect between performance and pay. Since salary does not appear to track performance rating, consider a formal compensation review to determine whether this is intentional (e.g., pay tied to role/tenure) or an area for policy improvement.
2. Review Florida's performance profile. Florida shows the highest count of “Poor” performance ratings of any region; a closer management review of that region's teams and processes may be warranted.
3. Examine the “Pending” status backlog. With 356 employees in Pending status — the largest single group — HR should confirm whether this reflects a genuine onboarding bottleneck or simply a data-maintenance gap.
4. Use the Department × Region heatmaps for workforce and real-estate planning. The clear concentration patterns (e.g., HR in Florida, DevOps in California) can directly inform office space, hiring, and resourcing decisions.
5. Continue tracking the 2023 hiring surge. Understanding what drove the jump to 228 hires in 2023 will help the client plan future headcount growth more predictably.

6. Maintain the cleaned dataset in the PostgreSQL database as the single source of truth. Future reporting, dashboards, or analyses should build on the employee_details table rather than reverting to the original raw file.

6. Conclusion

This engagement successfully converted a deliberately messy, partially incomplete employee dataset into a fully cleaned, structured, and database-ready resource, and used it to generate fourteen distinct visual analyses spanning performance, compensation, remote work, regional and departmental distribution, and hiring trends. Every cleaning decision in this report was made using transparent, defensible, and industry-standard methods, and every chart was generated directly from the final cleaned dataset — meaning the client can trust that the insights presented here are accurate, reproducible, and ready to support real business and HR decisions.

We recommend the client treat the cleaned dataset stored in the PostgreSQL database as the authoritative version going forward, and use the recommendations in Section 5 as a starting point for deeper follow-up analysis where appropriate.

End of Report